

Distinct Permutations with Identical Elements

MATH 797 — Week 1

Statement

One classic problem in combinatorics is counting combinations and permutations. However, if some of the elements in the permutation are identical, such as letters in a word, then we would have to bring out different methods to enumerate such permutations. Consider the word MISSISSIPPI. If the letters in this word were not identical and instead were different colors (red I, blue I, yellow I, etc.), then the number of distinct permutations would simply be $11!$. To account for the repeated letters, we divide by the number of permutations for each specific letter. For example, the letter S shows up 4 times so we divide by $4!$, divide by $2!$ for the two P's, divide by $4!$ for the four I's, and divide by $1!$ for the single M. Thus, we get: